

**EL Phase 2**



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# ELAPSE

## Entropy-Linked Autonomous Propagation and Self-Erasure

*A Multi-Domain Ensemble Framework for Active Data Containment in P2P Networks*

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### **Mathematical Modelling EL**

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# The Problem: Sensitive Data in P2P Networks



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Once sensitive data enters a P2P network, it replicates autonomously — reaching large fractions of nodes before any fixed timeout fires.

## Prior Work

- ✗ Vanish (key decay)
- ✗ Ephemerizer
- ✗ Fixed chronological timers

💡 ELAPSE Insight: Replace the chronological clock with Shannon entropy of the spatial data distribution as the erasure trigger — creating an adaptive, topology-sensitive deletion policy.

### Topology-Blind

Fixed timers unaware of network density & hub structure

### Sybil Vulnerable

Key-decay protocols suffer Sybil attacks (Wolchok et al.)

### Centralised Assumption

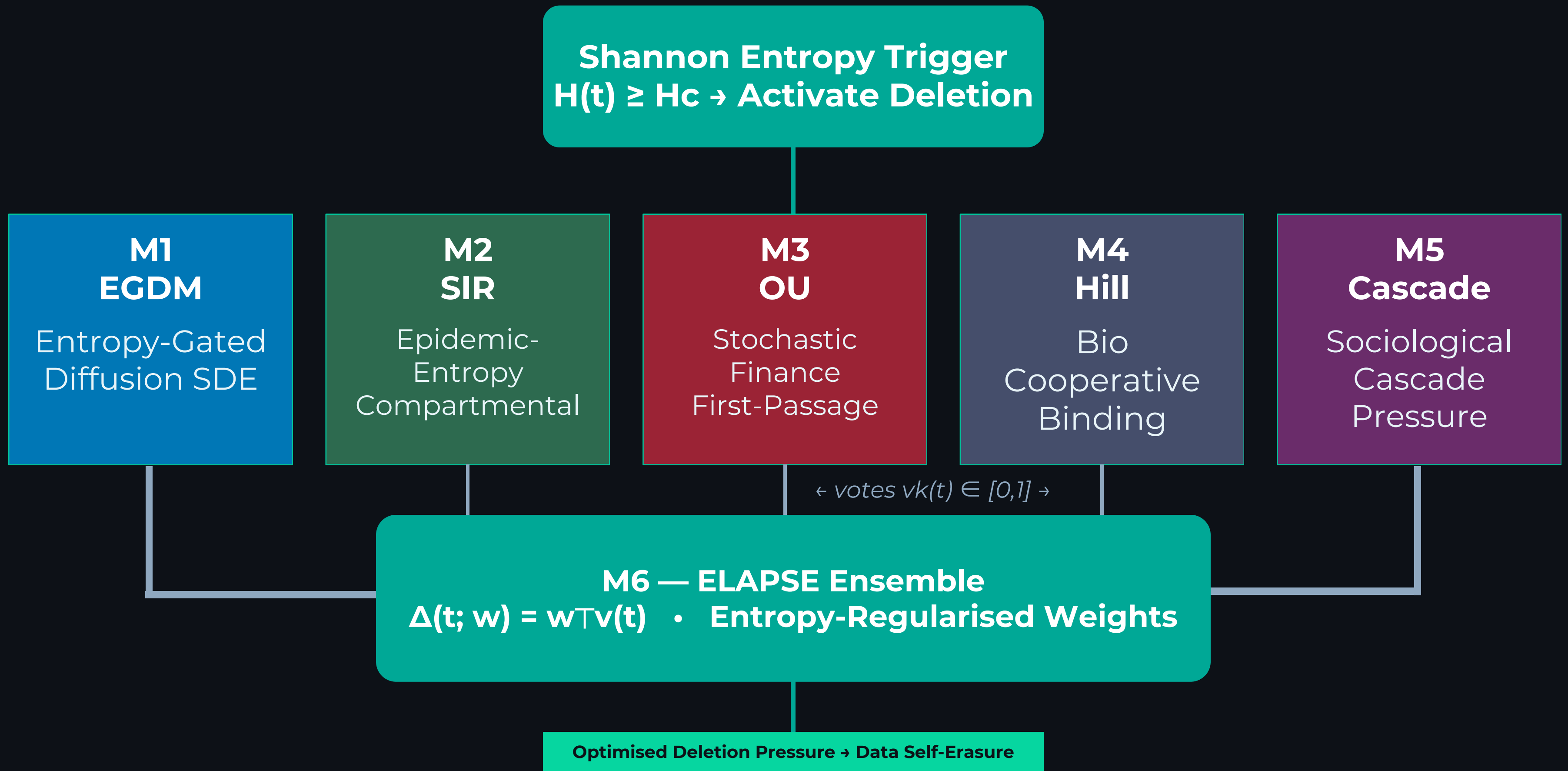
Machine unlearning requires central control

# ELAPSE Framework Architecture



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# The Five Deletion Mechanisms



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## M1 · Physics

### Entropy-Gated Diffusion (EGDM)

SDE-based data spread with deletion pressure. Vote rises polynomially above  $H_c$  threshold.

$$\begin{aligned} dx_i &= [-\alpha(Lx)_i - \mu\Delta x_i + s_i]dt + \sigma_n\sqrt{x_i} \\ dw_i \end{aligned}$$

## M2 · Epidemiology

### Epidemic-Entropy SIR

Data modelled as infectious load. Vote = mean infected fraction across nodes.

$$dI_i/dt = \beta S_i \sum A_{ij} I_j - \gamma I_i - \mu \Delta I_i$$

## M3 · Finance

### Stochastic First-Passage (OU)

Ornstein-Uhlenbeck barrier model. Vote derived from spectral entropy growth rate.

$$v_3(t) = 1 - \exp(-\lambda_2 \alpha t / (H_{\max} - H_c))$$

## M4 · Biology

### Cooperative Binding (Hill Function)

Sigmoidal switching from systems biology. Hill coefficient  $n_h=4$  gives stable transition.

$$v_4(t) = H(t)^{n_h} / (H_c^{n_h} + H(t)^{n_h})$$

## M5 · Sociology

### Sociological Cascade Pressure

Combines global entropy signal with local neighbourhood cascade threshold.

$$v_5 = \frac{1}{2}v_1 + \frac{1}{2} \cdot (\text{frac. nodes above cascade thresh } \kappa)$$

# Three Corrected Theoretical Results



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1

## Well-Posedness (Non-Circular Proof)

ELAPSE SDE has a unique, pathwise non-negative strong solution on  $[0, T]$ .

*Fix: Earlier proofs invoked global existence to prove non-negativity — circular. Now: Feller boundary classification first (entrance-not-exit), then global existence via Markov inequality.*

2

## Stochastic IEE Upper Bound

Bound extended from  $\sigma_n=0$  to the full stochastic system ( $\sigma_n > 0$ ).

*Fix: Second-moment Gronwall argument:  $\gamma = -2\mu\delta\min + \sigma_n^2 < 0$  at paper's parameters ( $\gamma = -0.15$ ), confirming average mass decay.*

3

## Itô-Corrected Spectral Decay

$\sigma_n^2$  correction enters quadratic variation of  $p_i = x_i/M$ , not its drift.

*Fix: Earlier version incorrectly attributed  $\sigma_n^2$  to the drift of  $dp_i$ . Numerically  $< 2\%$  of the main spectral term — trigger time  $\tau(\lambda_2)$  accurate to within 2%.*

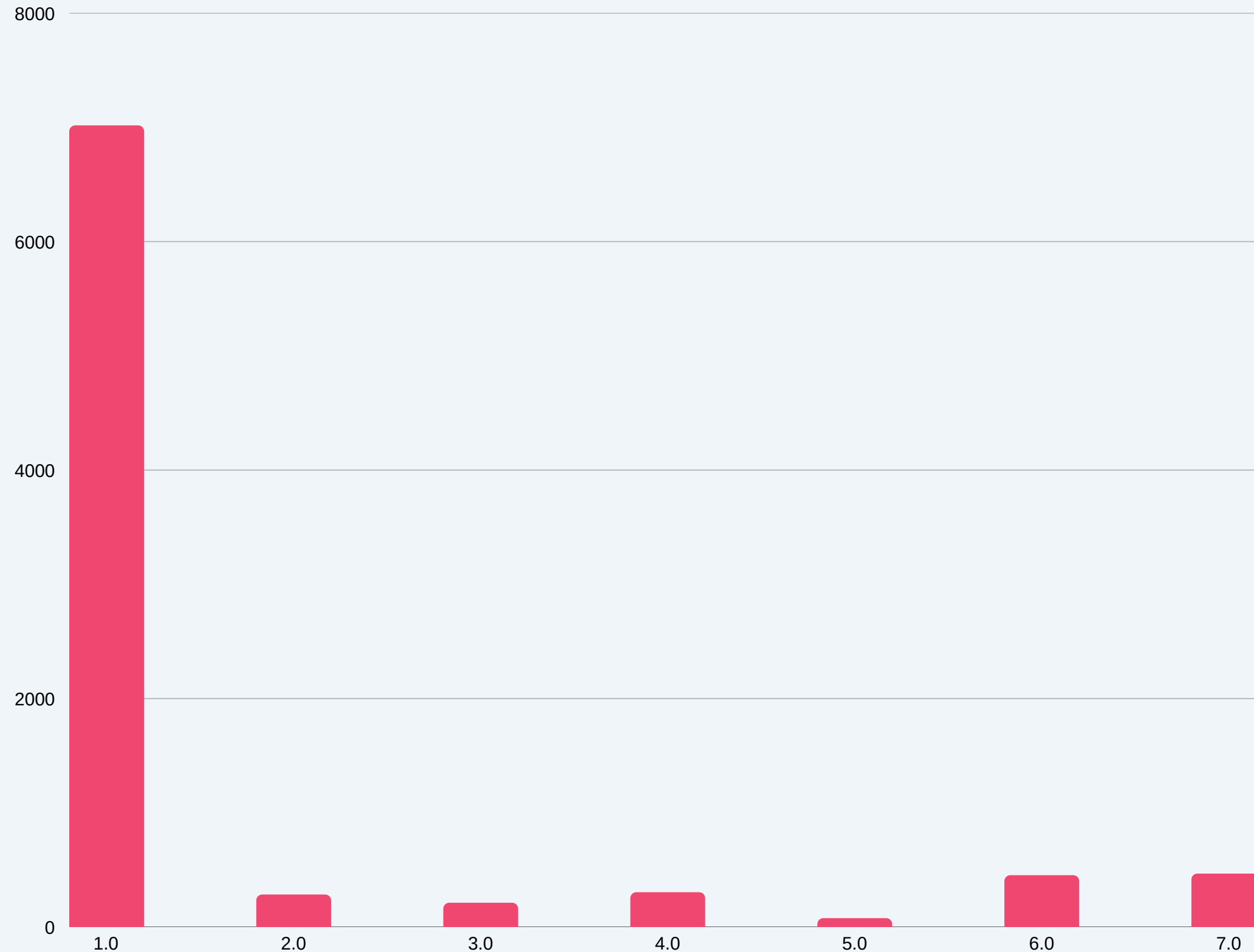


# Empirical Results: IEE Reduction at n = 500



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5-fold cross-validation, N=50 seeds, 95% CI

## >90%

IEE reduction  
all mechanisms vs M0

## 3.4%

M6 worst-case CV  
vs 6.9% for M5

## 91%

M6 lower IEE  
on evolving ER→BA

# M5 vs M6: Six-Metric Dominance Scorecard



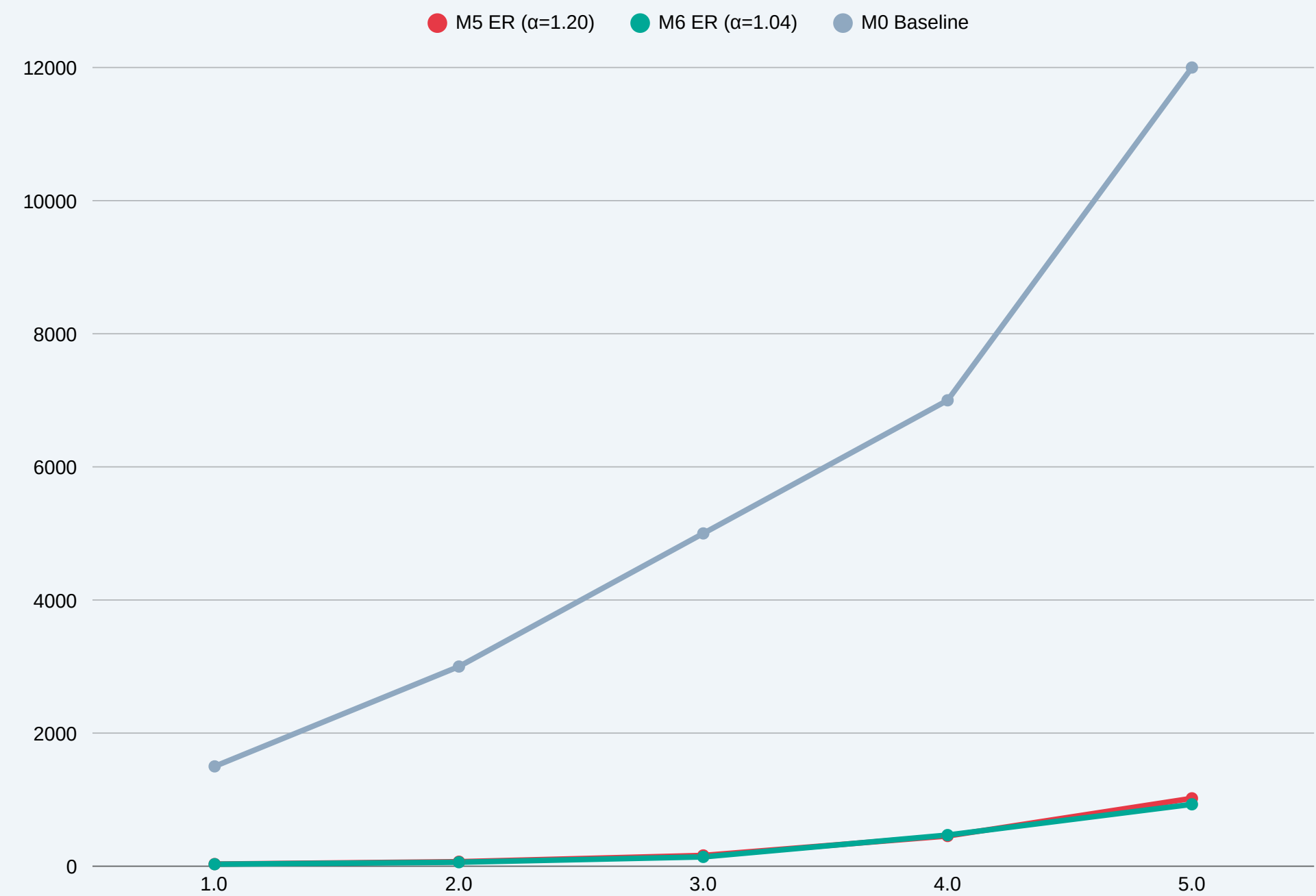
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| Metric                        | M5 Social | M6 (topo) | M6 (mixed) |
|-------------------------------|-----------|-----------|------------|
| Mean IEE (static, matched)    | 68.60 ✓   | 96.48     | 98.45      |
| Worst-case CV (%)             | 6.94      | 3.45 ✓    | 3.83 ✓     |
| Evolving-network IEE          | 1021.16   | 90.03 ✓   | 90.03 ✓    |
| Scaling exponent $\alpha$     | 1.25      | 1.03 ✓    | 1.05 ✓     |
| Deletion reliability CV (%)   | 5.17      | 3.14 ✓    | 3.24 ✓     |
| Cross-topo reliability CV (%) | 4.75      | 4.32 ✓    | 3.13 ✓     |

✓ = Best on this dimension     $n = 100$ ,  $N = 50$  seeds, Bonferroni-corrected  $t$ -tests ( $p < 0.001$ ,  $|d| > 6.9$ )

Use M6 when: topology is dynamic | graph scale >  $n^* \approx 9,832$  | consistent cross-seed performance is required

# Scaling Analysis & Real-World Validation



Crossover at  $n^* \approx 9,832$ : M6 achieves lower IEE than M5 above this scale

## Real-World Network Snapshots

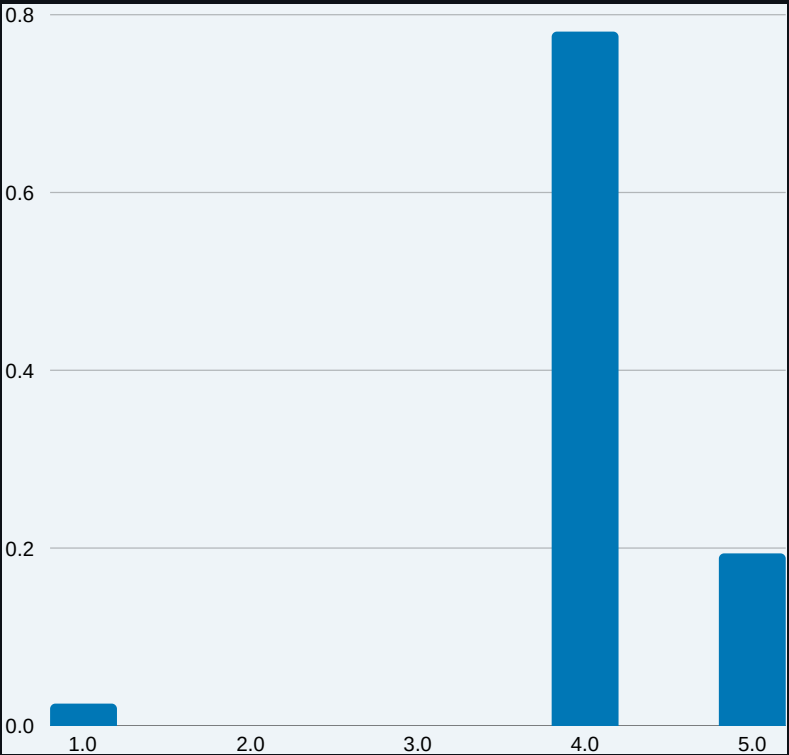
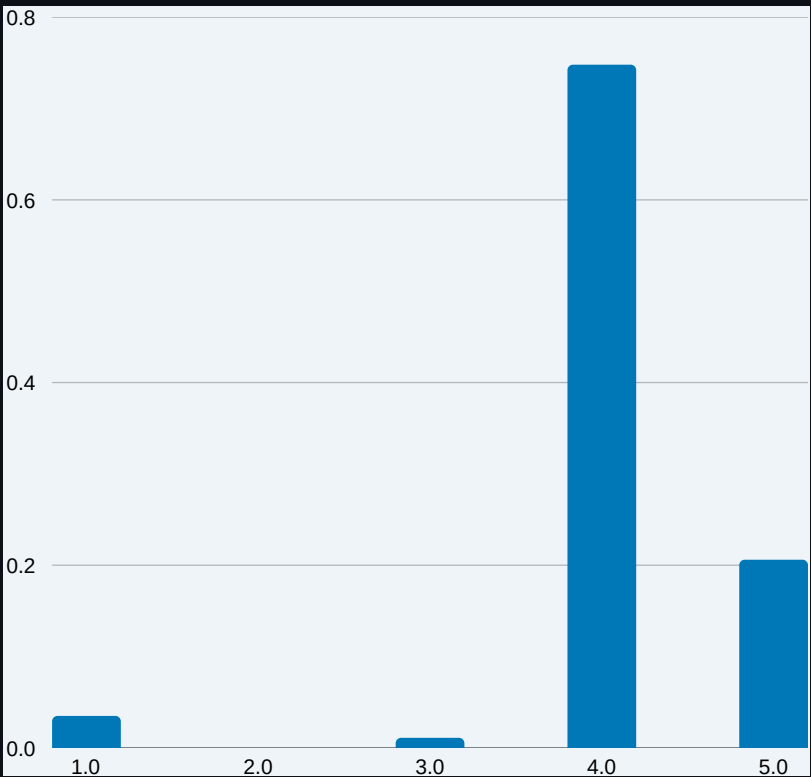
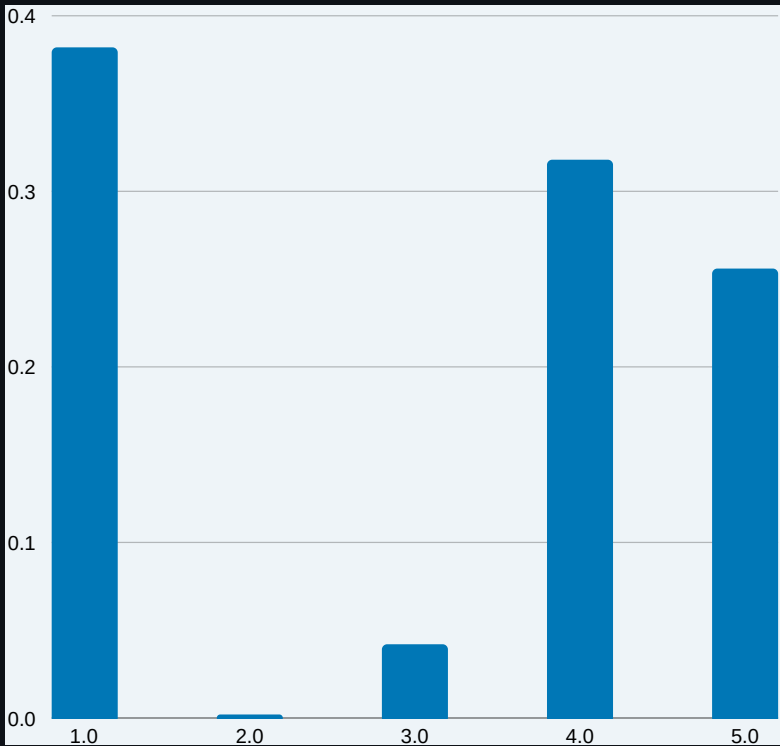
| Network        | n   | M0 IEE | M5 IEE | M6 IEE |
|----------------|-----|--------|--------|--------|
| Gnutella08 bfs | 500 | 6962   | 473    | 619    |
| Gnutella08 rnd | 71  | 640    | 63     | 75     |
| Gnutella31 bfs | 500 | 6782   | 534    | 670    |
| CA-GrQc        | 412 | 5409   | 388    | 512    |
| Power-Grid     | 374 | 4723   | 357    | 462    |

All mechanisms reduce IEE >90% vs M0

📌 Sharp activation at  $H_c$  is a finite-size crossover, NOT a universal phase transition ( $H^*_c/H_{max}$  shifts from 0.525→0.838 as  $n$  grows 50→200, ruling out universality).



# Ensemble Weights & Mechanism Diversity



Biology (M4) dominates BA & WS due to stable sigmoidal switching; EGDM + Social dominate ER's well-connected hub structure.

## Regularisation Trade-off (BA, n=200)

|                           |             |                 |           |
|---------------------------|-------------|-----------------|-----------|
| $\lambda_{\text{reg}}=0$  | Max wk=1.00 | Collapses to M4 | IEE=157.0 |
| $\lambda_{\text{reg}}=5$  | Max wk=0.81 | Some diversity  | IEE=165.2 |
| $\lambda_{\text{reg}}=15$ | Max wk=0.61 | Balanced ★      | IEE=174.9 |
| $\lambda_{\text{reg}}=20$ | Max wk=0.54 | Too spread      | IEE=179.3 |

Mixed-trained M6 generalises across all topologies — best worst-case IEE guarantee among M6 variants.

# Conclusions



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🔒 ELAPSE replaces fixed erasure timers with entropy-based triggering — topology-aware, adaptive, and physics-grounded.

📊 All 5 mechanisms reduce IEE by >90% over uncontrolled diffusion across ER, BA, and WS topologies.

🏆 M5 wins raw IEE on matched static graphs. M6 wins on reliability (CV 3.4% vs 6.9%), evolving networks (91% reduction), and large-scale crossover at  $n^* \approx 9,832$ .

✅ Three theoretical gaps closed: non-circular well-posedness, full stochastic IEE bound ( $\sigma_n > 0$ ), and Itô-corrected spectral decay.

🌐 Real-world validation on Gnutella, CA-GrQc, Power-Grid networks confirms >90% IEE reduction. Welch t-tests:  $p < 0.001$ ,  $|d| > 6.9$ .

Future Work: Closed-loop RL optimal control · Extensions to directed graphs · [github.com/Hitme02/ELAPSE](https://github.com/Hitme02/ELAPSE)